

Agentic IaC Automation

From Chat to Cloud in Minutes

-
- ✓ 5 specialized agents working together
 - ✓ Knowledge grounded in MCP Servers & Tools
 - ✓ Human-in-the-loop governance at every critical gate
 - ✓ \$1.29M–\$1.79M yearly savings per team < 1 month break-even

Agentic IaC Automation

Every cloud engagement hits the same bottleneck: IaC authoring requires deep cloud expertise, governance isn't enforced by default, and Day 2 operations remain largely manual with limited visibility. Reference architectures and proven patterns exist across the group but are not searchable or shared, so teams rebuild from scratch, producing code that is rarely reused, inconsistently structured, and siloed within individual engagements.

TerraBot is a multi-agent platform that transforms plain-English infrastructure requests into production-ready, standards-compliant Terraform code complete with a GitHub PR, CI/CD trigger, and live architecture diagram. This lowers the skills barrier so any engineer can self-serve infrastructure, while they retain full control over what gets deployed.

HOW

Functional Solution

Conversational IaC through a chat interface where engineers interact with TerraBot using natural language. Five agents — Code Generation, Standards & Policy, Security Scanning, PR Workflow, and Deploy Orchestration — are coordinated by an orchestrator that monitors each stage and feeds outputs back when issues are detected.

WHAT

Business Challenges

- IaC authoring requires deep cloud expertise, diverting high-cost resources from strategic work and delaying project delivery.
- Code and patterns are siloed per project, no reuse, no shared asset discoverability across teams.
- No enforced governance, inconsistent naming, structure, and tagging lead to compliance failures.
- Misconfigurations are caught late at plan/apply stage, causing costly rework.
- Terraform expertise is concentrated in a few senior engineers

Use Case Description

A Cloud Engineer describes an infrastructure need in plain English. TerraBot's five specialized agents orchestrate the end-to-end response: gathering requirements through targeted clarifying questions, validating the Azure subscription state, enforcing organizational naming, tagging, and security policies, generating modular Terraform HCL informed by live Registry docs and Azure Verified Modules via the Terraform MCP Server, and rendering an architecture diagram directly from the generated code. Key triggers are the initial engineer request, human approval at the code-review gate, and human approval at the terraform-plan gate, after which TerraBot opens a GitHub PR, monitors the CI/CD pipeline, and executes deployment.

Technical Solution

Built on the Microsoft Agent Framework and deployed as Foundry Hosted Agents. Agents follow a sequential maker/checker pipeline with the orchestrator checking and correcting outputs at each stage. The Foundry Model Router handles model selection automatically across all agents. Agents are tooled with Terraform, Azure, and GitHub MCP Servers for live provider schemas, subscription and state validation, and repository operations. Security scanning uses tfsec and Checkov, CI/CD runs through GitHub Actions, and identity is managed via Microsoft Entra ID with full audit trail.

WHO

Stakeholders

Cloud Platform Engineers
Solution Architects
Delivery Managers
DevOps Engineers
Capgemini Clients

WHY

KPIs Impacted

- Faster IaC authoring cycle
- Fewer security misconfigurations
- Reduced deployment lead time
- Faster release cycles
- Higher standards compliance rate
- Fewer PR rejections and rework loops

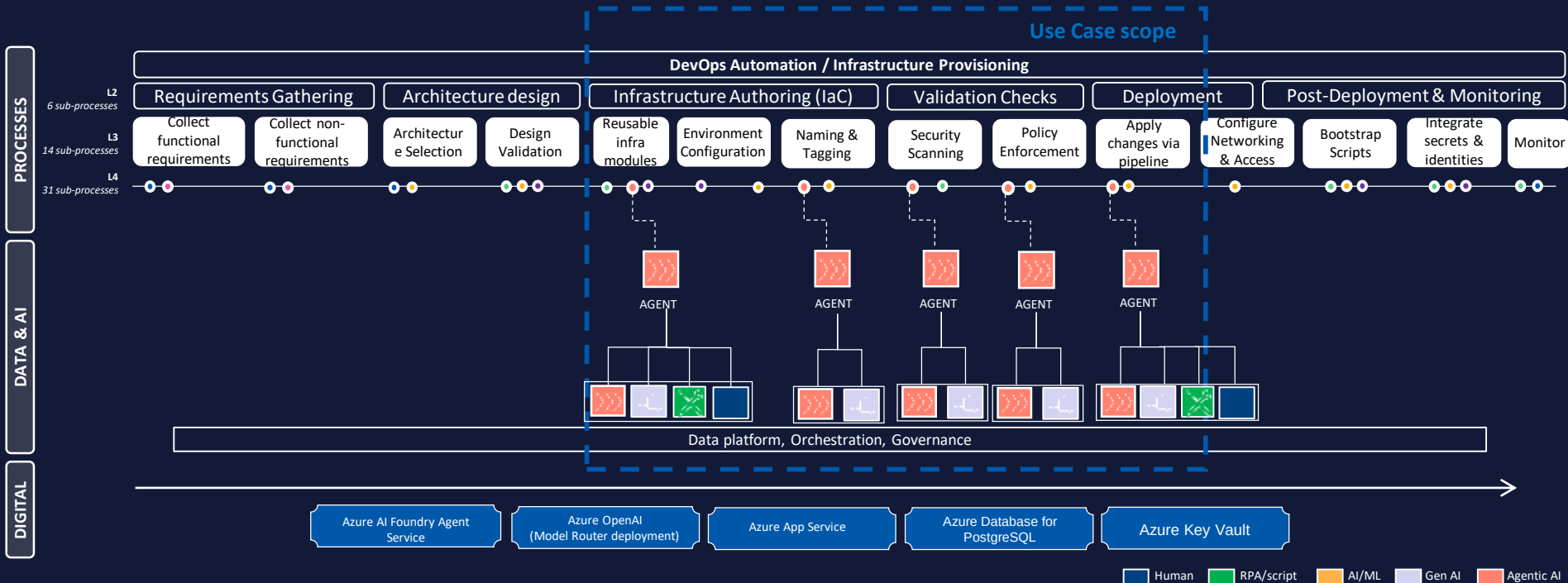
Business Impact

- Lower skills barrier for infrastructure provisioning
- Increased engineer capacity for higher-value work
- Reduced cost of compliance failures Less knowledge silos across delivery teams
- Faster onboarding of junior engineers to cloud projects



Agentic DevOps Automation

Use Case Scope in the Business Process



Outcomes

Faster Build

More Design Time

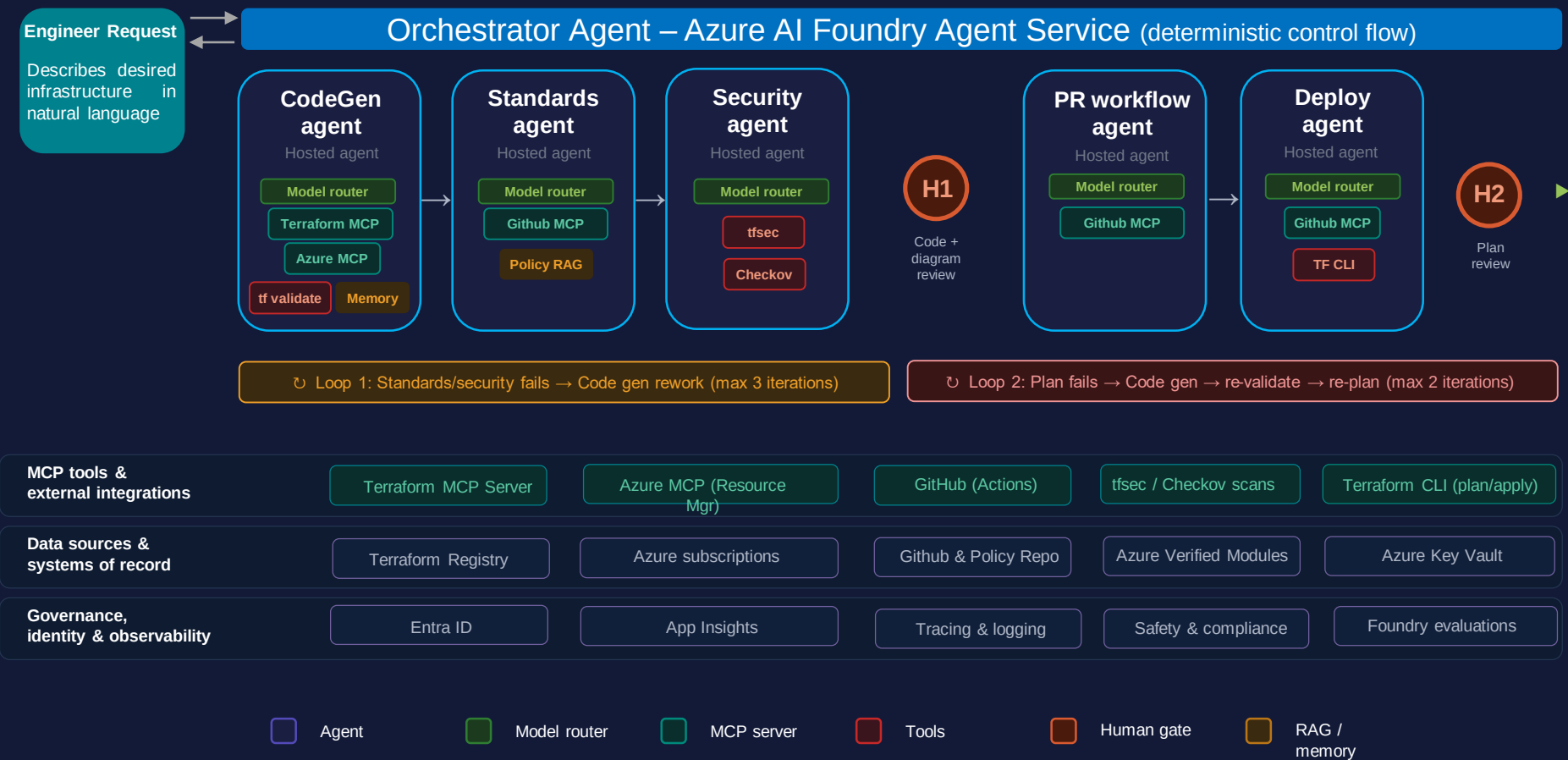
Auto Compliance

Productivity gain

Less Siloed Engagements & Teams

Less Time Spent on Operational Watch

Functional Solution



Technical Architecture



1 Presentation layer — Azure App Service (React / Next.js)

Chat interface
Conversational IaC

File explorer
Generated HCL

Pipeline dashboard
Plan/apply status

Diagram viewer
Architecture SVG

2 Orchestrator — Azure AI Foundry Agent Service

Foundry Agent Service
Sequential + maker-checker

Azure OpenAI
Model Router deployment

Foundry memory
Conversation state

3 Hosted agents (Foundry)

CodeGen agent
HCL generation

Standards agent
Policy checker

Security agent
tfsec + Checkov

PR workflow agent
GitHub Octokit

Deploy agent
Terraform CLI

4 MCP servers and tool integrations

Terraform MCP
Live registry schemas

Azure MCP
ARM Resource Mgr

GitHub
Actions + Octokit

tfsec / Checkov
Static analysis

Terraform CLI
Plan / apply

5 Data sources and systems of record

PostgreSQL
Azure DB (managed)

Terraform Registry
HashiCorp (external)

Azure subscriptions
Target environments

GitHub repos
VCS + policies

Key Vault
Secrets (zero in code)

6 Governance, identity & observability

Entra ID + RBAC
SSO, managed identity

Azure Key Vault
Zero secrets in code

App Insights
Metrics + alerting

Foundry tracing
Agent-level spans

Foundry evaluations
Quality scoring

SOC2 logging
No PII in logs

Terraform state
Customer-owned

VNet integration for backend services · Private endpoints for Key Vault + PostgreSQL · Managed identity (service-to-service) · No public admin endpoints



Presentation



Orchestration



Agents



MCP / tools



Data



Governance



Azure Service Selection

Azure AI Foundry Agent Service

Core

Managed agent runtime with multi-agent orchestration, built-in memory, and MCP support. Core orchestration layer.

Azure OpenAI (Model Router)

AI

Model Router analyses request complexity and routes automatically. No hardcoded models per agent.

Azure App Service

Compute

Hosts React/Next.js frontend and stateless Express backend. Horizontally scalable.

Azure Database for PostgreSQL

Data

Persistent storage for conversations, settings, generated files. Managed and encrypted at rest.

Azure Key Vault

Security

Encrypted storage for GitHub PATs, Azure credentials, and API keys. Zero secrets in code.

Microsoft Entra ID + RBAC

Security

Identity and access control. Managed Identity for service-to-service auth. No shared secrets.

App Insights + Foundry Tracing

Ops

Agent observability: latency, error rates, token usage, evaluation scores, agent-level spans.

VNet + Private Endpoints

Network

VNet integration for backend services. Private endpoints for Key Vault + PostgreSQL. No public admin.

Terraform MCP Server

MCP

Live provider schemas from HashiCorp Registry. Eliminates hallucinated resource arguments.

Azure MCP Server

MCP

ARM Resource Manager for subscription state validation and infrastructure discovery.

GitHub MCP Server

MCP

Repository operations, PR workflows, Actions pipeline monitoring via Octokit.

Foundry Evaluations

Ops

Quality scoring on agent outputs. Measures code quality, policy compliance, and accuracy.

tfsec

Tool

Static security analysis for Terraform code. Scans HCL for misconfigurations and vulnerabilities.

Checkov

Tool

Policy-as-code scanner for infrastructure. Catches compliance violations before deployment.

Terraform CLI

Tool

Handles init, validate, plan, and apply operations. Core deployment execution engine.

GitHub Actions

Deploy

CI/CD pipeline for Terraform plan/apply. PR-triggered workflows with human approval gates.



Terrabot's KPIs & Impact

- IaC authoring time accelerated
- Security misfiguration reduction
- Reduction of deployment time
- Faster release cycle

Business Impact

TerraBot **reduces** DevOps **overhead**, **downtime**, and compliance **risk**, unlocking **faster delivery** and **major cost savings**

BUSINESS CASE

Current Costs (without TerraBot)

DevOps/IT team (~5 engineers)	\$900k-\$1.2M/yr
Team labor overhead	\$600k-\$900k
Team strategy work	\$300k-\$600k
Azure IaC Services	<i>In sunk cost</i>
Downtime & compliance	~\$750K-\$1.2M/yr
Total Current Cost	~1.65M-\$2.4M/yr

New Costs (with TerraBot)

DevOps/IT team (~5 engineers)	\$400k-\$600k/yr
Implementation (one-time)	\$5k-\$25k
Azure IaC Services	<i>same</i>
Azure AI Foundry tokens + hosting	\$1.8k-\$9.6k/yr
Terrabot Maintenance	\$15k-\$24k/yr
Downtime & compliance	~\$110K-\$213K/year
Total New Cost (annual)	~\$500K-\$900K/yr

PAYBACK IN

<1 MONTH

via cost savings

ANNUAL LABOR SAVINGS

\$500K+

Per 5-person team per year

CLOUD OPTIMIZATION

\$150K+

In eliminating 15-20% of resource waste

Total Estimated Savings (annual)

\$1.29M-\$1.79M/yr

Hackathon Evaluation: 10 Use Case Criteria



LAYER 3: STRATEGIC VALUE

Problem Significance

Infrastructure automation **lags at just 56%**, IaC orchestration remains underutilized due to complexity and steep skills requirements.

User Impact

Empowers broader delivery teams to self-serve, significantly reducing **knowledge silos**.

KPI Impact

Delivers **faster release cycles**, fewer PR rejections, and higher standards compliance.

LAYER 2: AI ORCHESTRATOR ENGINE

Innovation

Multi-agent platform transforming requests into code, triggered pipelines, and **live diagrams**.

AI Fit

Combines code synthesis via the Foundry Model Router with **deterministic scanning tools**.

User Experience

Conversational chat interface requiring zero prior Terraform expertise to initiate.

Reusability (MCP)

Agents are seamlessly toolled with **Terraform, Azure, and GitHub MCP Servers**.

LAYER 1: ENTERPRISE FOUNDATION

Responsible Design

Strict **maker/checker pipeline** requiring Human Gate 1 (Code) and Gate 2 (Plan) approvals.

Feasibility

Built natively on the Microsoft Agent Framework as **Foundry Hosted Agents**.

Scalability & Security

Secured via **Microsoft Entra ID** with a full audit trail, alongside **Checkov and tfsec**.